# Huffman Code

Sh.1. Të kompresohet kodi i mëposhtëm duke përdorur Huffman Coding:

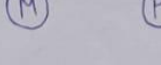
MISSISSIPPI

1° Konstruktimi i tabelës:

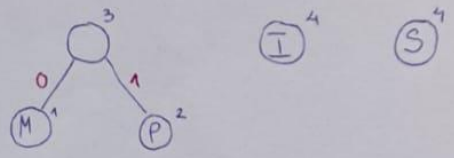
| char | frequency | encoding |
|---|---|---|
| I | 4 | 11 |
| M | 1 | 100 |
| P | 2 | 101 |
| S | 4 | 0 |

2° Konstruktimi i Trie (nga frekuenca më e vogël)

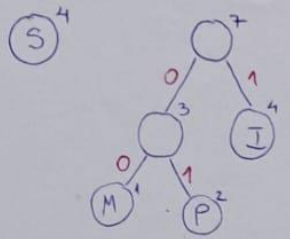
H.1°



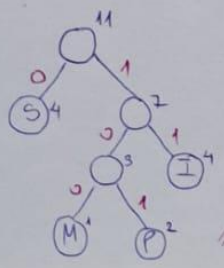

H.2°




H.3°




H.4°




* bitat e kodimit:

I: 11
M: 100
P: 101
S: 0

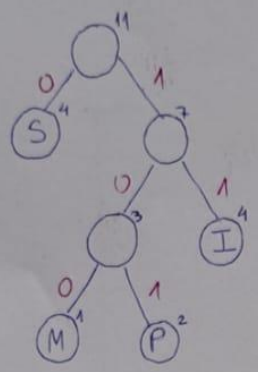
* kompresimi:

M I S S I S S I P P I
100 11 0 0 11 0 0 11 101 101 11

MISSISSIPPI i kompresuar

# Dekompresimi me Huffman

Të dekompresohet tekst i mëposhtem nëse eshte dhënë Trie:




100 11 00 11 00 11 101 101 11

M I SS I SS I P P I ← tekst i dekompresuar

Sh. 2. a) Të kompesohet :

ABRACADABRA !

1°

| char | freguency | encoding |
|---|---|---|
| ! | 1 | 1010 |
| A | 5 | 0 |
| B | 2 | 110 |
| C | 1 | 1011 |
| D | 1 | 100 |
| R | 2 | 111 |

Kompresim :

A B R A C A D A B R A !

↓ ↓ ↓ ↓ ↓ ↘ ↘ ↘ ↘ ↘ ↘

0 110 111 0 1011 0 100 0 110 1110 1010

2° Trie :

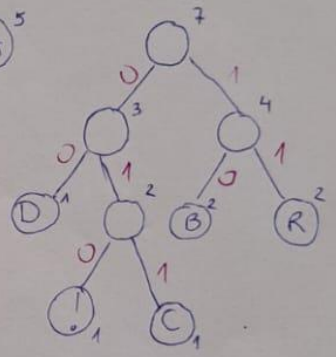
1° ! 1 C 1 D 1 B 2 R 2 A 5

2° D 1 (2: 0 → !, 1 → C) B 2 R 2 A 5

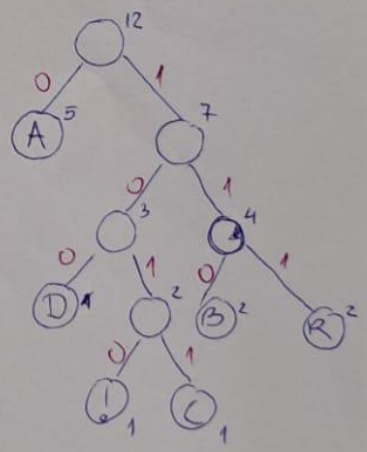
3° B 2 R 2 (3: 0 → D, 1 → 2: 0 → !, 1 → C) A 5

4° (3: 0 → D, 1 → 2: 0 → !, 1 → C) (4: 0 → B, 1 → R) A 5

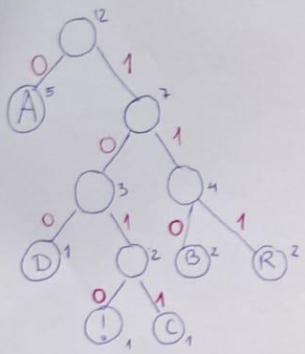
5° A 5 (7: 0 → 3: 0 → D, 1 → 2: 0 → !, 1 → C; 1 → 4: 0 → B, 1 → R)



6° (12: 0 → A, 1 → 7: 0 → 3: 0 → D, 1 → 2: 0 → !, 1 → C; 1 → 4: 0 → B, 1 → R)



b) Të dekompresohet:




0 110 111 0 1011 0 100 0 110 111 0 100

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓

A B R A C A D A B R A!

Detyrë: Të kompresohet tekstin "YELLOWWOODDOOR" duke përdorur Algoritmin Huffman.

Shn.1 Të kompresohet teksti :

# LZW - Kompresimi (dictionary based compression algo

M I S S I S S I P P I S S I P P I me algoritmin LZW

| key (current) | next | value (output) | dictionary |
|---|---|---|---|
| M | I | 2 | MI → 5 |
| I | S | 1 | IS → 6 |
| S | S | 4 | SS → 7 |
| S | I | 4 | SI → 8 |
| I | S | / | / |
| IS | S | 6 | ISS → 9 |
| S | I | / | / |
| SI | P | 8 | SIP → 10 |
| P | P | 3 | PP → 11 |
| P | I | 3 | PI → 12 |
| I | S | / | / |
| IS | S | / | / |
| ISS | I | 9 | ISSI → 13 |
| I | P | 1 | IP → 14 |
| P | P | / | / |
| PP | I | 11 | PPI → 15 |
| I | / | 1 | / |

(kompresimi)

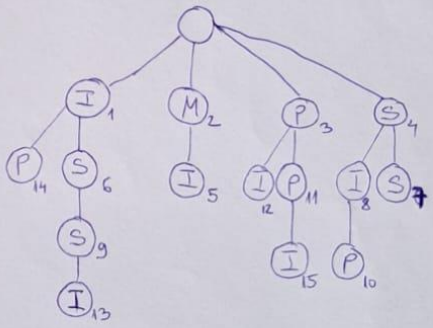
| key | value |
|---|---|
| I | 1 |
| M | 2 |
| P | 3 |
| S | 4 |

kompresimi :

M I S S I S S I P P I S S I P P I

2 1 4 4 6 8 3 3 9 1 11 1

tre :



1) LZW - Dekompresimi

2 1 4 4 6 8 3 3 9 1 11 1

| key | value |
|---|---|
| I | 1 |
| M | 2 |
| P | 3 |
| S | 4 |
| MI | 5 |
| IS | 6 |
| SS | 7 |
| SI | 8 |
| ISS | 9 |
| SIP | 10 |
| PP | 11 |
| PI | 12 |
| ISSI | 13 |
| IP | 14 |
| PPI | 15 |

# LZW - (with ASCII code)

Sh. 2. M I S S I S S I P P I

| key | value |
|---|---|
| M | 77 |
| I | 73 |
| S | 83 |
| P | 80 |

| key | next | value | dictionary |
|---|---|---|---|
| M | I | 77 | MI → 257 |
| I | S | 73 | IS → 258 |
| S | S | 83 | SS → 259 |
| S | I | 83 | SI → 260 |
| I | S | / | / |
| IS | S | 258 | ISS → 261 |
| S | I | / | / |
| SI | P | 260 | SIP → 262 |
| P | P | 80 | PP → 263 |
| P | I | 80 | PI → 264 |
| I | / | 73 | / |

Kompresimi : M I S S I S S I P P I

77 73 83 83 258 260 80 80 73

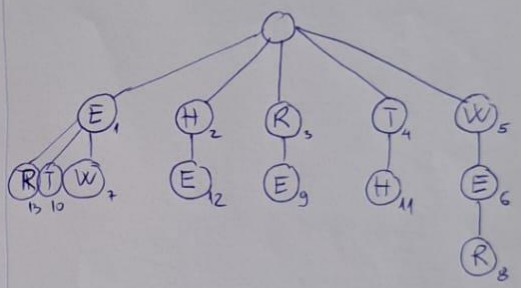
Sh.3. Të kompresohet teksti :

W E WERE T H E R E

| key | next | value | dictionary |
|---|---|---|---|
| W | E | 5 | WE → 6 |
| E | W | 1 | EW → 7 |
| W | E | / | / |
| WE | R | 6 | WER → 8 |
| R | E | 3 | RE → 9 |
| E | T | 1 | ET → 10 |
| T | H | 4 | TH → 11 |
| H | E | 2 | HE → 12 |
| E | R | 1 | ER → 13 |
| R | E | / | / |
| RE | / | 9 | / |

| key | value |
|---|---|
| E | 1 |
| H | 2 |
| R | 3 |
| T | 4 |
| W | 5 |

Trie :




kompresimi :

W E W E R E T H E R E

5 1 6 3 1 4 2 1 9

Sh.3. Të dekompresohet:

| 5 | 1 | 6 | 3 | 1 | 4 | 2 | 1 | 9 |
|---|---|---|---|---|---|---|---|---|
| W | E | WE | R | E | T | H | E | RE |

| 5 | 1 | 6 | 3 | 1 | 4 | 2 | 1 | 9 |
|---|---|---|---|---|---|---|---|---|
| W | E | WE | R | E | T | H | E | RE |

| key | value |
|---|---|
| E | 1 |
| H | 2 |
| R | 3 |
| T | 4 |
| W | 5 |
| WE | 6 |
| EW | 7 |
| WER | 8 |
| RE | 9 |
| ET | 10 |
| TH | 11 |
| HE | 12 |
| ER | 13 |

Detyrë: Të kompresohet teksti: POSSESSIVENESSNESSSSIVENESS duke përdorur algoritmin LZW.